AI / ML

# ML Training Pipelines

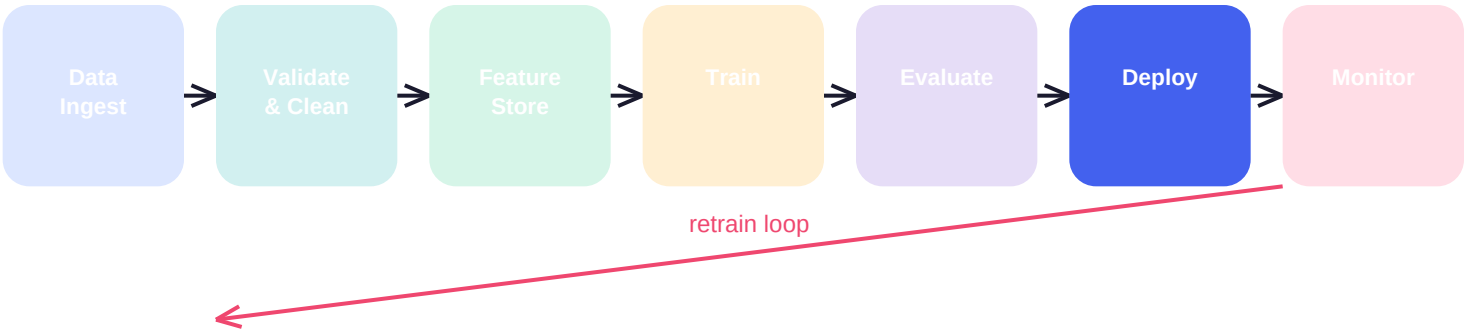
From raw data to deployed model:
the production machine-learning lifecycle.

**ANIMATED WALKTHROUGH**
Step through pages to play the animation

STEP 1/6

# The Pipeline at a Glance

ML systems are software systems — with data as a first-class citizen.




Monitoring closes the loop — drift triggers retraining, not a pager at 3am.

STEP 2/6

# Data Validation: Garbage In, Garbage Out

Validate data like you validate code.

### Schema

Columns, types, ranges match expectations

### Distribution

No sudden shifts in feature stats (mean, nulls, cardinality)

### Anomalies

Outliers quarantined, not silently trained on

### Lineage

Every row traceable to its source + transform version

Tools: Great Expectations, TFDV. A broken upstream column shouldn't silently poison your model.

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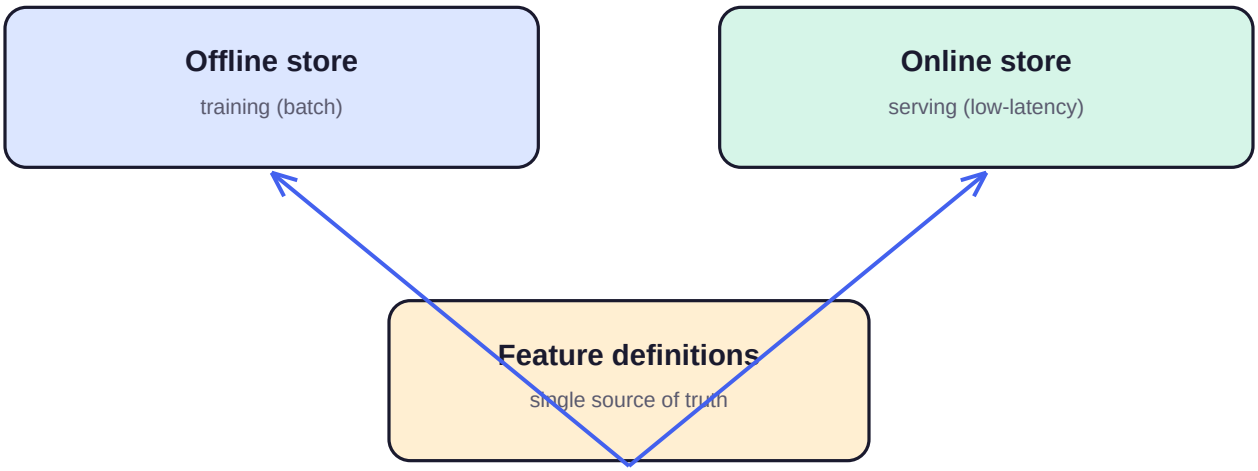
# Feature Stores: One Definition of 'User Age'

Train and serve must compute features IDENTICALLY — or you get skew.




**Training-serving skew kills models silently**

Same code, same features, both paths. Point-in-time correctness for training.

STEP 4/6

# Experiment Tracking

If it isn't logged, it didn't happen.

Log per run: code commit • data version • hyperparams • metrics • artifacts

| Run | LR | Batch | Val Acc |
|---|---|---|---|
| a3f9 | 3e-4 | 256 | 0.941 |
| b7c1 | 1e-4 | 512 | 0.937 |
| d2e8 | 3e-4 | 512 | 0.948 ✓ |

Tools: MLflow, Weights & Biases. Compare apples to apples — same data split, same metric.

STEP 5/6

# Deployment Patterns

Getting the model to users — safely.

### Batch inference

Score millions of rows overnight. Simple, cheap, high latency.

### Real-time API

Model behind REST/gRPC. Milliseconds matter; autoscale it.

### Shadow mode

New model scores live traffic silently — compare before promoting.

### Canary

5% of traffic to v2; watch metrics; roll forward or back.

STEP 6/6

# Monitoring: Models Rot

The world changes; your model doesn't — unless you watch.

### Data drift

Input distribution shifts (new users, seasonality). Detect with PSI/KS tests.

### Concept drift

The RULES change (fraud patterns evolve). Labels needed to detect.

### Feedback loops

Model's outputs become its future inputs — bias amplifies.

### SLOs for ML

Latency, throughput, prediction quality — alert like any service.

STEP ✓I✓

# Key Takeaways

## ML Pipelines — Remember

- Treat data as code: version it, validate it, track its lineage.
- Feature stores kill training-serving skew — one definition, two stores.
- Log every experiment: code + data + params + metrics, or it didn't happen.
- Deploy progressively: shadow, canary, then full rollout.
- Monitor for drift and decay — retraining is a loop, not an event.

**Keep exploring**

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev